
Learning the Geometry of Political Discourse through Contrastive Rhetorical Deformations

EL BAZ Avner*

Abstract

Quantifying rhetorical phenomena such as *langue de bois* remains challenging, as they are not reducible to lexical frequency or local syntactic patterns but instead emerge from global discourse organization. Classical lexical approaches offer interpretability but fail to capture the structural and distributional nature of political language.

In this work, we revisit this problem through contrastive representation learning. We introduce a Multi-Contrastive Noise Contrastive Estimation (NCE) framework operating on chunked document representations, where discourse structure is learned through controlled transformations between direct and rhetorically constrained formulations. This allows us to study political discourse as an induced geometry in embedding space rather than a set of annotated categories.

We further compare this approach with a lexical baseline and show that contrastive representations exhibit a structured latent organization across political groups, with stable separation patterns and robustness to demographic variables. We analyze this space through two complementary functionals: a max-based score capturing extreme alignment under rhetorical deformation, and a profile-based score capturing the distributional structure of constraint activations.

Overall, the results suggest that contrastive learning induces a non-isotropic discourse geometry where rhetorical variation is encoded as structured deformation rather than lexical variation. This provides a quantitative framework for studying political language as a structured transformation process rather than a purely symbolic object.

Introduction

Institutional communication is rarely a direct encoding of factual content. In high-stakes environments, language is often shaped by constraints that prioritize coordination, legitimacy, and risk management over explicit representation. A canonical example is the Challenger disaster in 1986, where engineering concerns regarding O-ring failure under low temperatures were progressively reframed within managerial discourse, ultimately shifting technical uncertainty toward operational justification [on the Space Shuttle Challenger Accident \[1986\]](#). This case illustrates how institutional language can systematically transform risk into acceptable narratives.

A similar tension appears in political discourse, where statements must simultaneously encode information, maintain institutional positioning, and remain publicly acceptable. Prior work in discourse analysis has linked these dynamics to framing effects, strategic ambiguity, and ideological structuring rather than purely informational communication [Entman \[1993\]](#), [Chong and Druckman \[2007\]](#). In this context, French political discourse of the late twentieth century exhibits a particularly salient phenomenon commonly referred to as *langue de bois*, characterized by abstraction, depersonalization, and controlled semantic opacity [Botezatu Tilița \[2019\]](#).

*Corresponding author: FRANCOIS Gabriel

Importantly, *langue de bois* is not well captured as a discrete linguistic category. Instead, it can be interpreted as a distributional property emerging from systematic patterns of formulation, omission, and reframing. Recent work in computational linguistics suggests that such properties are better modeled in continuous representation spaces than through surface-level lexical statistics [Rodriguez et al. \[2023\]](#), [Ma and Wen \[2023\]](#).

This limitation is particularly evident for lexical approaches, which rely on token frequencies or predefined dictionaries. While interpretable, these methods fail to capture structural regularities distributed across entire documents, especially when equivalent rhetorical functions are expressed through heterogeneous surface forms. As a result, they conflate lexical overlap with semantic similarity and overlook discourse-level organization.

Representation learning offers a natural alternative by embedding texts into continuous semantic spaces where similarity is induced geometrically. In particular, contrastive learning frameworks such as InfoNCE have become central for learning sentence representations by explicitly structuring positive and negative pairs in embedding space [Chen et al. \[2020\]](#), [Jaiswal et al. \[2020\]](#). These methods replace hand-crafted similarity metrics with data-driven geometric constraints.

In this work, we operationalize *langue de bois* as a measurable property of such learned representations. We compare a lexical baseline with a contrastive representation learning framework based on Noise Contrastive Estimation (NCE), operating over chunked discourse segments. The contrastive signal is constructed from controlled transformations between direct formulations and rhetorically constrained variants of political statements, enabling explicit modeling of stylistic deformation.

This setup induces a Multi-Contrastive learning objective in which discourse structure is learned through alignment and opposition between semantically equivalent but rhetorically distinct formulations. Rather than treating discourse as a bag of independent features, we analyze it as a structured deformation of meaning in representation space.

Our contributions are threefold: (i) we construct a dataset of political discourse enriched with controlled rhetorical transformations, (ii) we introduce a contrastive retrieval framework for modeling discourse variation, and (iii) we analyze the induced embedding space to uncover systematic geometric structure across political groups.

All data, code, and experiments are publicly available on [GitHub](#), ensuring full reproducibility. We also acknowledge Gabriel FRANCOIS for his complementary analysis based on the POLAR framework, which provided additional insight into the structural properties of the dataset.

1 Dataset construction and preprocessing

The dataset construction follows a dual-corpus design separating real political discourse from a synthetically generated contrastive structure. This distinction is central to the framework, as the learning signal does not emerge directly from the real corpus but from controlled transformations defined in a separate generation pipeline. The primary corpus consists of 12,746 French political documents collected from electoral manifestos, campaign declarations, and institutional speeches spanning the period 1981-1993. After preprocessing and metadata filtering, 12,497 documents are retained. Each document is associated with structured socio-political attributes, including candidate identity, electoral context, and political affiliation. Standard normalization procedures are applied, followed by segmentation into overlapping chunks of approximately 450 tokens with a stride of 120 tokens, ensuring compatibility with transformer-based encoding while preserving local discourse coherence [Jaiswal et al. \[2020\]](#).

Importantly, this corpus is not directly used to construct contrastive pairs, but rather serves as the evaluation domain on which learned representations are applied.

In parallel, a second dataset is generated to model rhetorical transformations explicitly. A base set of political statements is first produced using controlled prompts grounded in historical actors, situations, and topics, enforcing stylistic constraints consistent with French political discourse from the 1980s-1990s.

Each generated sentence is then transformed into a contrastive pair through a constraint-driven paraphrasing process, producing aligned direct and rhetorically altered (*langue de bois*) formulations

while preserving factual content. Transformations are guided by structured constraint families capturing intentional, structural, and informational modifications of discourse.

The resulting synthetic dataset is organized as groups of contrastive patterns rather than independent samples. Each group defines a specific configuration of rhetorical transformations, forming a localized axis of variation between direct and constrained discourse.

These groups are used to define the training signal in a Noise Contrastive Estimation framework, where representation learning is driven by the separation of positive and negative examples in embedding space. Unlike standard contrastive setups that rely on implicit augmentations, the present approach leverages explicitly controlled linguistic transformations as the source of contrast.

This design introduces a strict functional separation between the two datasets: the real corpus provides naturally occurring discourse to be analyzed, while the synthetic corpus defines the transformation space used to learn representations. As a result, *langue de bois* is modeled not as a directly observed feature, but as a structured deformation captured through contrastive learning.

2 From lexical similarity to retrieval-based representation learning

Defining similarity between political texts is not straightforward. In institutional discourse, semantically close statements may share little surface overlap, while conversely, near-identical lexical realizations can correspond to distinct communicative functions. This weak coupling between form and meaning complicates any attempt to rely solely on surface-level similarity measures.

A first and historically dominant approximation models texts through lexical overlap. In this setting, documents are represented as bags of normalized lexical units, typically obtained via lemmatization to reduce morphological variation, a standard practice in information retrieval and NLP pipelines [Jurafsky and Martin \[2023\]](#). This representation provides a stable canonical form across inflections and improves comparability between documents, but it remains fundamentally local in nature.

Building on this representation, a natural baseline consists in dictionary-based scoring schemes, widely used in institutional text analysis. Each document is assigned a score based on the proportion of lemmas belonging to a predefined lexicon W :

$$\text{Score}_d = \frac{\sum_{i=1}^{L_d} \mathbf{1}(l_i \in W)}{L_d} \times 100$$

where L_d denotes the number of lemmas in document d . This formulation has the advantage of interpretability, as it directly quantifies lexical alignment with a targeted vocabulary. However, it also introduces a strong simplifying assumption: that meaning can be approximated through local lexical matches, independently of syntactic structure or discourse-level organization.

This assumption becomes particularly fragile in political language. Meaning is often not anchored in specific lexical items, but distributed across patterns of formulation, reformulation, and rhetorical positioning. As a consequence, two statements may express equivalent intentions while sharing no significant lexical overlap, whereas highly similar lexical content may correspond to different pragmatic functions. In this setting, lexical similarity tends to entangle surface form with semantic function, limiting its ability to capture paraphrastic variation and stylistic deformation associated with phenomena such as *langue de bois* [Botezatu Tilița \[2019\]](#).

Rather than discarding this limitation entirely, it motivates a shift in perspective. Instead of treating similarity as an explicit function over discrete tokens, recent approaches embed texts into continuous vector spaces where similarity is induced by the geometry of learned representations [Levy et al. \[2015\]](#). Within this framework, similarity is no longer predefined but learned as a property of the embedding space itself, typically through retrieval-based training objectives [Reimers and Gurevych \[2019\]](#).

A common formalization of this idea relies on contrastive objectives such as InfoNCE, which structure representation spaces by distinguishing between positive and negative pairs [Chen et al. \[2020\]](#). Importantly, this does not eliminate the notion of similarity, but rather shifts its definition from symbolic overlap to learned invariance under perturbations of the input space.

In this work, we extend this perspective by grounding representation learning in controlled discourse transformations rather than uncontrolled lexical variation. Instead of comparing raw documents

directly, we construct explicit pairs of semantically aligned but stylistically divergent sentences, contrasting direct formulations with rhetorically constrained variants while preserving factual content [Rodriguez et al. \[2023\]](#).

From this standpoint, representation learning does not replace lexical similarity but partially redefines it: similarity is no longer tied to token overlap, but to invariance under structured rhetorical transformations. This allows us to separate stylistic variation from factual content and to study political discourse as a space organized by controlled deformations rather than purely lexical proximity.

3 NCE-guided retrieval as contrastive discourse structure modeling

Classical retrieval formulations define similarity as a nearest-neighbor problem in a continuous embedding space, where the objective is to retrieve texts closest to a query under a geometric metric. While effective, this formulation remains limited for political discourse, where meaning is not only expressed through proximity but also through systematic transformations of linguistic form.

Rather than retrieving isolated similar texts, we focus on structured transformations that preserve factual content while altering rhetorical expression. This leads to a shift from instance-level retrieval to structure-level retrieval, where the basic unit of analysis is no longer a single document but a contrastive group of transformations.

Each group g_i is defined as:

$$g_i = (\mathbf{P}_i, \mathbf{N}_i)$$

where $\mathbf{P}_i$ corresponds to direct formulations and $\mathbf{N}_i$ to controlled rhetorical transformations. Given an encoder $h(\cdot)$, all textual elements are embedded in a shared representation space:

$$q = h(c), \quad \mathbf{P}_i = \{h(p_{i,j})\}, \quad \mathbf{N}_i = \{h(n_{i,j})\}$$

We define a contrastive objective inspired by Noise Contrastive Estimation (NCE):

$$\mathcal{L}(q, g_i) = -\log \frac{\sum_{p \in \mathbf{P}_i} \exp(\text{sim}(q, p)/\tau)}{\sum_{z \in \mathbf{P}_i \cup \mathbf{N}_i} \exp(\text{sim}(q, z)/\tau)}$$

This score measures whether a representation aligns more strongly with the direct formulation space than with its rhetorical transformation, and provides the fundamental signal on which all subsequent analyses are built.

From this base quantity, we consider two complementary aggregation regimes.

The first focuses on extreme sensitivity to rhetorical transformations. For a given query, we take the maximum contrastive response across all groups:

$$\text{Score}^{\max}(q) = \max_{g_i \in \mathcal{G}} \mathcal{L}(q, g_i)$$

and define the document-level version as the average of these maximal responses across chunks:

$$\text{Score}_D^{\max} = \frac{1}{n} \sum_{j=1}^n \text{Score}^{\max}(h(c_j))$$

This score captures the worst-case instability of a document under controlled rhetorical perturbations, and is used in the empirical analysis of latent structure.

The second aggregation regime is introduced for linguistic profiling and is grounded in the set of retrieved transformation groups:

$$\mathcal{R}_K(q) = \text{TopK}_{g_i \in \mathcal{G}} \mathcal{L}(q, g_i)$$

Rather than aggregating contrastive losses directly, we reinterpret $\mathcal{R}_K(q)$ as inducing a structured distribution over constraint occurrences, shaped by the selected transformation groups.

In this setting, we decompose the retrieved set into three disjoint subsets corresponding respectively to structural, conceptual, and informational constraints.

$$\mathcal{R}_K(q) = \mathcal{S}_K \cup \mathcal{C}_K \cup \mathcal{I}_K$$

Each element $g \in \mathcal{R}_K(q)$ belongs to exactly one of these subsets, i.e. $g \in \{\mathcal{S}_K, \mathcal{C}_K, \mathcal{I}_K\}$. For a given constraint λ_{i_0} , we define its normalized frequency within a group g as:

$$f_g(\lambda_{i_0}) = \frac{\text{card}(\lambda_{i_0} \in g)}{\text{card}(g)}$$

This quantity provides a local estimate of constraint activation within each transformation group, effectively characterizing the internal composition of g . We then define the query-level representation as the aggregation of these group-wise profiles over the full partition:

$$p(q) = \sum_{g \in \mathcal{S}_K \cup \mathcal{C}_K \cup \mathcal{I}_K} f_g$$

Since $\{\mathcal{S}_K, \mathcal{C}_K, \mathcal{I}_K\}$ forms a partition of $\mathcal{R}_K(q)$, this aggregation corresponds to summing the frequency profiles induced by all retrieved groups.

At the chunk level, we propagate this representation via:

$$\mathbf{p}(c) = \mathbf{p}(h(c))$$

The document-level profile score is then computed by averaging over all chunks:

$$\text{Score}_D^{\text{prof}} = \frac{1}{n} \sum_{j=1}^n \mathbf{p}(c_j)$$

This score captures the expected distribution of constraint types induced by the dominant transformation structures associated with a document. Importantly, this aggregation operates at the document level only, since the constraint taxonomy $\{S, C, I\}$ is already internalized within each retrieved group.

In contrast to the max-based score, which emphasizes sensitivity to the most discriminative transformation, this profile-based measure captures the global structural composition of constraint activations across retrieval-induced regimes.

Taken together, these two functionals capture complementary geometric regimes of the same contrastive space. The max-based score isolates extreme alignment directions, reflecting worst-case sensitivity to local rhetorical deformation, while the profile-based score aggregates distributional mass and encodes the stable organizational structure of constraint activations induced by the retrieval space.

They do not describe independent properties, but instead correspond to two complementary views of a single latent geometry: one driven by rare yet highly discriminative directions, and the other governed by the persistent structure of typical configurations.

The resulting framework characterizes discourse simultaneously through its instability under extreme contrast and its coherent organization across dominant transformation regimes.

4 Empirical structure of InfoNCE scores and latent discourse geometry

We analyze the empirical structure induced by the InfoNCE representation space with a focus on its most stable and informative regularities. Rather than enumerating isolated statistics, the objective is to characterize the global organization of the latent space and the way it reflects political discourse structure.

A first and central observation is that the learned representation space is not homogeneous. It exhibits a clear and persistent organization aligned with political groupings, suggesting that the contrastive objective induces a structured geometry rather than a simple semantic similarity space.

When aggregating scores at the level of political affiliation, a robust structural separation emerges across groups. This separation is not driven by a few outliers but reflects a consistent pattern observed

across the full dataset of 12,746 documents. As illustrated in Figure 1, political families occupy distinct regions of the score distribution: some groups, such as the Front National and the PCF, are systematically associated with higher values of the aggregated InfoNCE signal, while Ecologistes and Independent actors tend to concentrate in lower regions of the latent space. This ordering is stable and reproducible across the dataset, indicating that the learned representation encodes a meaningful alignment between political affiliation and latent positioning.

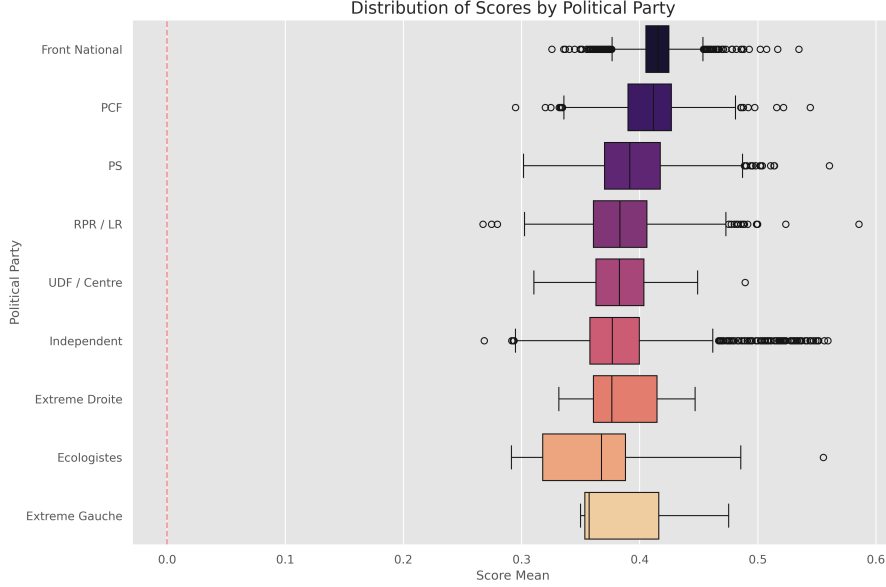


Figure 1: Distribution of InfoNCE scores across political parties. The latent space exhibits a structured organization aligned with political affiliations, revealing non-isotropic geometry.

Beyond differences in central tendency, the internal shape of the distributions also carries important information. Some political groups are characterized by compact and concentrated score distributions, suggesting a high degree of internal coherence in their discourse patterns, whereas others exhibit significantly broader dispersion, reflecting greater heterogeneity in rhetorical expression. The latent space therefore does not only separate groups, but also encodes differences in their internal structural consistency.

This organization remains stable over time. The dataset spans more than a decade, from 1981 to 1993, a period marked by substantial political and institutional change, yet the global geometry of the representation space does not undergo major reconfiguration. As shown in Figure 2, average scores evolve smoothly while preserving the relative ordering of political groups. Minor fluctuations exist at yearly resolution, but they do not alter the overall structure, suggesting that the representation captures persistent properties of political discourse rather than transient lexical artifacts.

The stability of this geometry is particularly important in the context of contrastive learning, as it indicates that the InfoNCE objective induces representations that are robust to contextual variation and reflect deeper discourse organization rather than surface-level shifts.

In the following analysis, this structure is quantified using the aggregated maximum score per document, denoted $\text{Score}_D^{\max}$, which captures the strongest alignment of each document with its most discriminative contrastive transformation group. This measure is then used as the primary signal to characterize cross-party structure, temporal evolution, and latent separation patterns in the representation space.

These results indicate that the learned representation space exhibits a structured geometry in which political groups are separable and consistently organized along stable axes, with systematic differences in dispersion and temporal stability. The resulting space can be interpreted as a non-isotropic embedding shaped by contrastive learning, where $\text{Score}_D^{\max}$ serves as an aggregate measure of a document’s position within this structure.

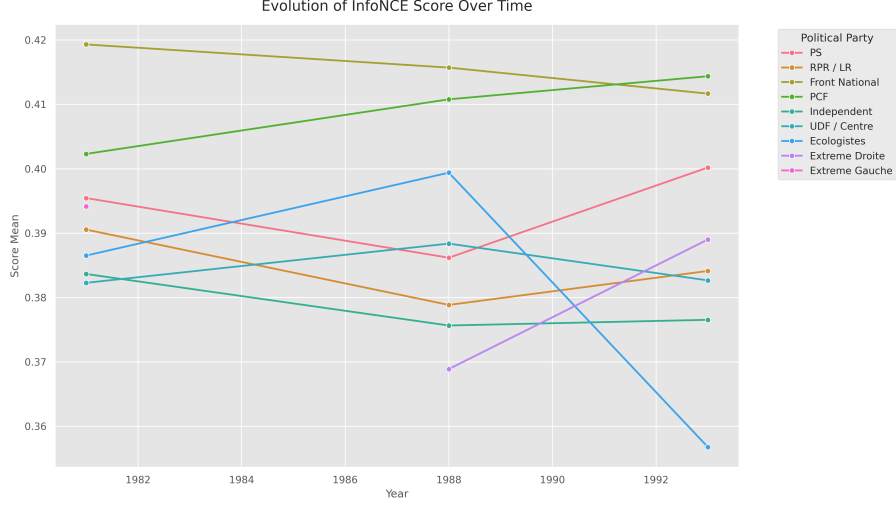


Figure 2: Temporal evolution of average InfoNCE scores by political group (1981-1993). The latent structure remains globally stable across time, with only minor fluctuations.

5 Analysis of InfoNCE Profiles and Relative Structure to the Global Mean

We build on the previously defined profile-based score $\text{Score}_D^{\text{prof}}$, which captures a distribution over constraint activations induced by the retrieval space. This quantity is treated as a random variable over documents, enabling a probabilistic interpretation of linguistic profiles at the corpus level.

At the first level of analysis, we consider conditioning this score on political parties Π . The corresponding group-specific profile is defined as:

$$\mathbb{E}_{\Pi} \left[\text{Score}_D^{\text{prof}} \right]$$

which represents the expected constraint activation profile induced by the retrieval structure within each party.

This defines the first representation used in the analysis, where each political party is characterized by its conditional expectation over the profile distribution. The second reference point is given by the global expectation over the full dataset:

$$\mathbb{E} \left[\text{Score}_D^{\text{prof}} \right]$$

The comparison between these two quantities yields a relative formulation:

$$\Delta_{\Pi} = \mathbb{E}_{\Pi} \left[\text{Score}_D^{\text{prof}} \right] - \mathbb{E} \left[\text{Score}_D^{\text{prof}} \right]$$

which captures how each political group deviates from the global linguistic baseline induced by the InfoNCE retrieval space.

To move beyond this scalar view, we lift the representation to a vectorial formulation. Each document is described by a constraint vector $\mathbf{c}$ encoding multiple families of linguistic properties, including informational, structural, and discourse-related constraints. This allows us to reinterpret $\text{Score}_D^{\text{prof}}$ as inducing a multidimensional profile space rather than a single aggregated statistic.

In this setting, the global reference profile becomes:

$$\mathbf{E}_{\text{global}} = \mathbb{E}[\mathbf{c}]$$

while each party-level profile is defined as:

$$\mathbf{E}_{\Pi} = \mathbb{E}[\mathbf{c} \mid \Pi]$$

The corresponding deviations are given by:

$$\Delta_{\Pi} = \mathbf{E}_{\Pi} - \mathbf{E}_{global}$$

As shown in Figure 3, these deviations cannot be reduced to uniform scalar shifts. Instead, they correspond to structured directional deformations in the constraint space, where each political party exhibits a coherent configuration of variations across multiple dimensions.

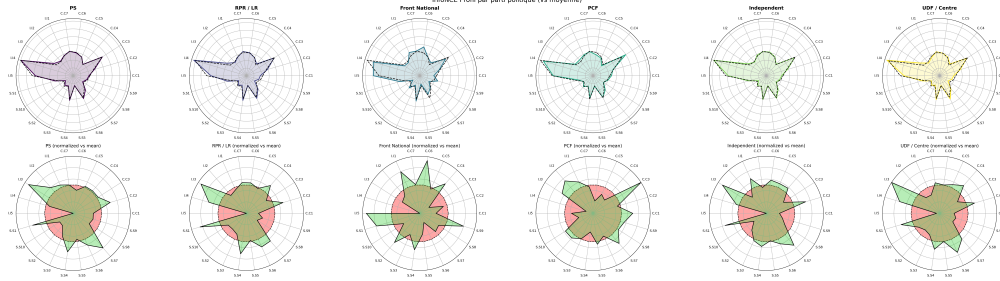


Figure 3: Linguistic profiles by political group: conditional expectations and deviations from the global mean profile.

More precisely, group differences manifest as systematically organized variations across constraint dimensions. For instance, the Front National exhibits a joint increase in informational fragmentation ($I.I5 = 0.0355$) and technical vocabulary usage ($C.C6 = +0.0353$), combined with a strong decrease in structural explicitness ($I.I4 = -0.0408$). This corresponds to a region of the latent space characterized by lexically dense yet structurally less articulated discourse.

In contrast, the PCF shows a different configuration, with a marked positive deviation on $C.C3$ ($+0.0271$), indicating stronger conceptual structuring, while remaining relatively stable on fragmentation-related dimensions. This reflects a more compact and internally organized discursive regime.

Intermediate groups such as the PS or Independents remain closer to the global mean while still exhibiting consistent micro-deformations. The PS, for example, shows a moderate increase in structural dimensions such as $I.I3$ ($+0.0169$), offset by a decrease in $I.I5$ (-0.0197), suggesting a more structured but still balanced configuration.

Overall, these patterns reveal a strong anisotropy in the profile space. Constraint dimensions do not vary independently but instead form stable co-variation structures that define coherent directional signatures of linguistic behavior.

From this perspective, $\mathbf{E}_{global}$ should be interpreted not merely as a statistical average, but as a geometric reference point induced by the contrastive learning dynamics. Each political party is then characterized by a displacement vector Δ_{Π} , encoding a structured deformation of this global discursive baseline.

Consequently, linguistic profiles are better understood not as static descriptors, but as directional trajectories within a constrained latent geometry. The InfoNCE framework thus induces a space organized around systematic transformations of a shared linguistic structure, rather than a flat similarity-based representation.

Robustness, Biases, and Generalization

We evaluate the robustness of the proposed framework across dataset regimes and modeling choices, focusing on whether observed effects reflect structural properties of the embedding space or artefacts of data construction.

The first source of potential bias lies in the manual design of rhetorical transformations. Although this introduces an explicit inductive bias between direct and constrained formulations, the induced effect remains consistent across settings. In both paraphrased and constrained regimes, LDB variants

systematically exhibit higher embedding-space dispersion than their corresponding base formulations, with gaps of +0.143 and +0.057 respectively. While the magnitude of this difference varies with the generation regime, its direction is preserved, indicating that rhetorical transformation acts as a stable deformation operator rather than a dataset-specific artefact.

We further assess whether this behavior depends on the underlying data distribution by comparing paraphrased and constrained corpora. Although absolute diversity increases in the constrained regime for both base and LDB representations (base: +0.196, LDB: +0.111), the relative ordering between discourse types remains unchanged. This suggests that dataset-level variation primarily induces a global rescaling of the embedding geometry, without altering its internal relational structure.

A complementary analysis is provided by the unconstrained LDB baseline. In this setting, LDB representations take intermediate values between paraphrased and constrained regimes, differing only slightly from the paraphrased case (+0.017). This indicates that constraint injection has limited impact in the absence of structured paraphrasing, and that most of the observed structure arises from the interaction between discourse transformation and representation learning rather than from constraints alone.

We also examine the stability of estimates across samples. The variance of all reported metrics remains low across conditions, suggesting that observed differences are not driven by stochastic fluctuations but reflect consistent properties of the embedding space at corpus scale.

Finally, cross-regime comparisons reveal a clear separation of effects: shifts induced by changes in corpus construction are larger than those induced by rhetorical transformation within a fixed corpus. However, these shifts preserve the relative ordering between base and LDB representations. This decoupling between magnitude and ranking indicates that the learned representation encodes a stable geometric deformation, where transformations act primarily by scaling rather than by restructuring the space.

Overall, these results support a notion of conditional robustness. While absolute values depend on dataset composition, three properties remain invariant: (i) the direction of the effect induced by rhetorical transformation, (ii) the ordering between base and LDB regimes, and (iii) the relative structure of embeddings across political groups. The framework therefore captures a deformation-invariant geometry of discourse, modulated in scale by dataset conditions but stable in structure.

Conclusion

This work introduces a contrastive framework for modeling political discourse as a structured geometric space rather than a collection of lexical patterns. We reinterpret rhetorical phenomena such as *langue de bois* as systematic deformations of meaning within a learned representation space.

Across experiments, the InfoNCE objective induces a strongly anisotropic embedding geometry in which political groups occupy stable and separable regions. This organization reflects not only differences in central tendency but also in dispersion and sensitivity to rhetorical transformation, indicating that discourse structure is encoded at multiple geometric scales.

Two complementary functionals characterize this space. The first captures extreme alignment under transformation, highlighting directions of maximal sensitivity to rhetorical deformation. The second aggregates constraint activations, revealing stable distributional organization across discourse regimes. Together, they describe a latent space structured both by rare discriminative directions and by persistent global organization.

This geometry remains stable across time despite substantial variation in the corpus, suggesting that the learned representation captures persistent structural properties of political discourse rather than transient lexical variation. Minimal influence from demographic variables further supports the interpretation that the embedding space primarily encodes discursive structure.

Robustness analyses across paraphrased, constrained, and unconstrained regimes confirm that the induced structure is not tied to a specific data pipeline. While dataset composition affects the scale of embeddings, it does not alter the relative ordering of discourse types or the qualitative nature of the induced geometry.

Overall, contrastive learning does not merely produce a similarity space, but induces a structured manifold in which rhetorical regimes correspond to controlled deformations of a shared linguistic baseline. Political discourse is thus better understood as a space of transformations acting on a common representational structure rather than as a set of discrete categories.

Future work should investigate whether this geometric structure is invariant across encoder architectures, replace hand-crafted constraint systems with learned latent factors, and extend the framework to longitudinal settings in order to model rhetorical evolution as trajectories in representation space.

References

- Ana-Maria Botezatu Tilița. La langue de bois dans le discours politique. *Anadiss*, 27:219–223, 2019. doi: 10.17684/issn.2393-1140. URL [https : //www.diacronia.ro/en/indexing/details/A29342](https://www.diacronia.ro/en/indexing/details/A29342).
- Ting Chen, Simon Kornblith, Mohammad Norouzi, and Geoffrey Hinton. A simple framework for contrastive learning of visual representations. In *Proceedings of the 37th International Conference on Machine Learning (ICML)*, 2020. URL [https : //arxiv.org/abs/2002.05709](https://arxiv.org/abs/2002.05709).
- Dennis Chong and James N. Druckman. Framing theory. *Annual Review of Political Science*, 10: 103–126, 2007.
- Robert M. Entman. Framing: Toward clarification of a fractured paradigm. *Journal of Communication*, 43(4):51–58, 1993.
- Ashish Jaiswal et al. A survey on contrastive self-supervised learning, 2020. URL [https : //arxiv.org/abs/2011.00362](https://arxiv.org/abs/2011.00362).
- Daniel Jurafsky and James H. Martin. *Speech and Language Processing*. 2023. URL [https : //web.stanford.edu/~jurafsky/slp3/](https://web.stanford.edu/~jurafsky/slp3/). Draft of 3rd edition.
- Omer Levy, Yoav Goldberg, and Ido Dagan. Improving distributional similarity with lessons learned from word embeddings. *Transactions of the Association for Computational Linguistics*, 3:211–225, 2015.
- Qian Ma and Qiufang Wen. Can space be conceptualized in a different way in political discourse? *Discourse and Society*, 2023. doi: 10.1177/09579265231188956. URL [https : //doi.org/10.1177/09579265231188956](https://doi.org/10.1177/09579265231188956).
- Presidential Commission on the Space Shuttle Challenger Accident. Report of the presidential commission on the space shuttle challenger accident, 1986. URL [https : //ntrs.nasa.gov/citations/19860019504](https://ntrs.nasa.gov/citations/19860019504).
- Nils Reimers and Iryna Gurevych. Sentence-bert: Sentence embeddings using siamese bert-networks. In *Proceedings of EMNLP*, 2019.
- Pedro L. Rodriguez, Arthur Spirling, and Brandon M. Stewart. Embedding regression: Models for context-specific description and inference. *American Political Science Review*, 2023. doi: 10.1017/S0003055422001228. URL [https : //doi.org/10.1017/S0003055422001228](https://doi.org/10.1017/S0003055422001228).

A Prompts

A.1 Generator Prompt

```
GEN_PROMPT = ""
Tu appartiens à {actors} en France entre 1980 et 1999.

Tu prends la parole lors de {situations}.

Tu énonces UNE phrase issue d'un discours politique sur {topics}.

RÈGLE ABSOLUE :
Réponds UNIQUEMENT en JSON valide.

-----
CONTRAINTES
-----

- une seule phrase
- maximum 30 mots
Au moins 50% des phrases un des éléments suivants :
- un nombre (date, volume, montant)
- un lieu précis (ville, région)
- une institution nommée
-----
STYLE
-----

- ton institutionnel
- vocabulaire concret
- formulation crédible pour les années 80-90
- éviter toute tournure générique répétée

-----
FORMAT STRICT
-----

{{
  "sentences": ["...", "..."]
}}
""
```

A.2 Paraphizer Prompt

```
PARA_PROMPT = ""
Tu es un générateur de dataset NLP contrastif basé sur des transformations linguistiques contrôlées.

RÈGLE ABSOLUE :
Tu réponds UNIQUEMENT en JSON valide.
Aucun texte hors JSON.

=====
HIÉRARCHIE DES RÈGLES (PRIORITÉ DÉCROISSANTE)
=====

RÈGLE 1 (PRIORITÉ ABSOLUE) :
Tu DOIS produire une opposition claire et marquée :
- LDB = langue de bois évidente (dilution, euphémisation, abstraction)
```

- DIRECT = formulation directe, explicite, voire plus brutale

Si l'opposition est faible, la sortie est invalide.

RÈGLE 2 :

Tu DOIS appliquer TOUTES les contraintes associées à chaque mode.

INTERDICTION ABSOLUE :

- ne jamais reformuler une contrainte
- ne jamais inventer de contrainte
- ne jamais fusionner des contraintes

Chaque contrainte doit être recopiée exactement comme fournie.

IMPORTANT :

Les contraintes ne sont PAS binaires.

Elles doivent être appliquées avec une INTENSITÉ ajustable.

- Si possible → application forte
- Si conflit avec RÈGLE 1 → application atténuée mais visible
- Interdit : ignorer complètement une contrainte

RÈGLE 3 :

Ne jamais modifier le fait (contenu informationnel strictement identique).

=====
OBJECTIF
=====

À partir d'une phrase factuelle, générer EXACTEMENT 1 paire (LDB / DIRECT).

=====
PRINCIPE
=====

- LDB = transformation langue de bois
- DIRECT = transformation directe

=====
ENTRÉES
=====

PHRASE SOURCE :
{sentence}

CONSTRAINTES :
{constraints}

=====
RÈGLES D'APPLICATION
=====

1. Générer LDB avec un style langue de bois maximal
2. Générer DIRECT avec un style frontal maximal
3. Appliquer toutes les contraintes :
 - Priorité : maximiser leur présence
 - Ajuster leur intensité si nécessaire
4. Ne jamais modifier le fait

```
=====
VALIDATION
=====
```

Invalide si :

- opposition faible entre LDB et DIRECT
- perte ou ajout d'information
- contrainte absente ou non perceptible

Valide si :

- toutes les contraintes sont présentes
- certaines contraintes sont atténuées mais détectables
- opposition forte maintenue

```
=====
FORMAT DE SORTIE STRICT
=====
```

```
{{
  "ldb": "...",
  "direct": "...
}}
```

```
=====
RÈGLE FINALE
=====
```

JSON strict uniquement.
Aucune explication.
Aucun texte hors structure.

PHRASE SOURCE :

```
{sentence}
"""
```

B Constraints

B.1 Constraint Generator

```
ACTORS = [
    "le gouvernement",
    "le Premier ministre",
    "le président de la République",
    "le ministre de l'économie",
    "le ministre du travail",
    "le ministre de l'intérieur",
    "le ministre de la justice",
    "le ministre de la santé",
    "le ministre de l'éducation",
    "le ministre de l'industrie",
    "le ministre du budget",
    "un ministre",
    "un secrétaire d'État",
    "le préfet",
    "un préfet de région",
    "le maire",
    "un maire de grande ville",
    "un maire de commune rurale",
```

```

"le conseil municipal",
"le conseil régional",
"le conseil général",
"l'administration centrale",
"les services de l'État",
"une administration locale",
"le Parlement",
"l'Assemblée nationale",
"le Sénat",
"une commission parlementaire",
"un rapporteur",
"un haut fonctionnaire"
]

SITUATIONS = [
    "une déclaration publique",
    "une intervention télévisée",
    "une session parlementaire",
    "une séance à l'Assemblée nationale",
    "une séance au Sénat",
    "une conférence de presse",
    "une allocution officielle",
    "une réunion interministérielle",
    "un conseil des ministres",
    "une visite officielle",
    "un déplacement en région",
    "une inauguration",
    "une réunion de crise",
    "un débat parlementaire",
    "une audition parlementaire",
    "une interview radiophonique",
    "une intervention médiatique",
    "une présentation budgétaire",
    "une discussion de loi",
    "une commission d'enquête",
    "une réunion administrative",
    "une rencontre avec des élus",
    "une réunion avec des partenaires sociaux",
    "une table ronde",
    "un sommet européen",
    "une négociation européenne",
    "une déclaration internationale",
    "une réunion diplomatique"
]

TOPICS = [
    "le chômage de masse",
    "la montée du chômage des jeunes",
    "les contrats précaires",
    "le développement du temps partiel",
    "la désindustrialisation",
    "la fermeture de sites industriels",
    "les délocalisations",
    "la compétitivité industrielle",
    "la modernisation industrielle",
    "les privatisations d'entreprises publiques",
    "les nationalisations",
    "la restructuration du secteur public",
    "la réforme de l'État",

```


"la modernisation administrative",
"le déficit public",
"la dette de l'État",
"la maîtrise des dépenses publiques",
"la fiscalité des ménages",
"la pression fiscale",
"la création de la CSG",
"le financement de la sécurité sociale",
"l'assurance maladie",
"le système de retraites",
"la réforme des retraites",
"le pouvoir d'achat",
"l'inflation",
"le logement social",
"la crise du logement",
"la politique de la ville",
"les banlieues",
"les violences urbaines",
"la délinquance",
"la sécurité intérieure",
"le maintien de l'ordre",
"la police nationale",
"la justice pénale",
"la réforme judiciaire",
"les prisons",
"l'immigration",
"le contrôle des frontières",
"le droit d'asile",
"l'intégration",
"les tensions identitaires",
"les discriminations",
"le racisme",
"la montée du Front national",
"la crise politique",
"l'abstention électorale",
"la cohabitation",
"le rôle du président",
"le rôle du gouvernement",
"la Ve République",
"la construction européenne",
"le traité de Maastricht",
"l'intégration européenne",
"la souveraineté nationale",
"la convergence économique",
"la monnaie européenne",
"les accords de Schengen",
"la libre circulation",
"l'éducation publique",
"la réforme scolaire",
"les universités",
"l'échec scolaire",
"les hôpitaux publics",
"le système de santé",
"les dépenses de santé",
"l'énergie nucléaire",
"EDF et GDF",
"les transports publics",
"la SNCF",
"les télécommunications",

```

    "la poste",
    "l'aménagement du territoire"
]

```

B.2 Constraint Paraphraser

```

CONSTRAINTS = {
  "INTENTION": {
    "C1": {
      "name": "euphémisation",
      "effect": "remplacer un terme direct ou négatif
                par une formulation atténuée ou neutre"
    },
    "D1": {
      "name": "explicitation",
      "effect": "remplacer toute atténuation par un terme direct et explicite"
    },

    "C2": {
      "name": "effacement_du_sujet",
      "effect": "supprimer ou masquer le sujet
                (passif, impersonnel)"
    },
    "D2": {
      "name": "sujet_explicite",
      "effect": "placer un sujet clair et identifié en début de phrase"
    },

    "C3": {
      "name": "flou",
      "effect": "rendre au moins un élément imprécis
                (temps, lieu, quantité, acteur)"
    },
    "D3": {
      "name": "precision",
      "effect": "rendre tous les éléments explicites et non ambigus"
    },

    "C4": {
      "name": "abstraction",
      "effect": "remplacer une action concrète par une formulation générique ou abstraite"
    },
    "D4": {
      "name": "concret",
      "effect": "utiliser des actions et objets concrets et observables"
    },

    "C5": {
      "name": "dilution_responsabilite",
      "effect": "éviter d'assigner clairement une responsabilité
                (passif, neutre)"
    },
    "D5": {
      "name": "responsabilite_claire",
      "effect": "attribuer explicitement l'action à un acteur identifié"
    },
  },
}

```

```

"C6": {
  "name": "jargon",
  "effect": "introduire vocabulaire administratif, technique ou bureaucratique"
},
"D6": {
  "name": "langage_simple",
  "effect": "utiliser un vocabulaire simple et courant"
},

"C7": {
  "name": "generalisation",
  "effect": "remplacer un élément précis par une formulation vague"
},
"D7": {
  "name": "specification",
  "effect": "remplacer les termes vagues par des éléments précis"
}
},
"STRUCTURE": {
  "S1": {
    "name": "phrase_longue",
    "effect": "phrase > 15 mots avec au moins une subordination
(ex: que, afin de, dans la mesure où)"
  },
  "R1": {
    "name": "phrase_courte",
    "effect": "phrase < 12 mots sans subordination,
structure simple sujet-verbe-objet"
  },
  "S2": {
    "name": "passif",
    "effect": "voix passive avec auxiliaire être + participe passé, sujet non acteur"
  },
  "R2": {
    "name": "actif",
    "effect": "voix active, sujet explicite en position initiale qui réalise l'action"
  },
  "S3": {
    "name": "nominalisation",
    "effect": "remplacement des verbes par noms abstraits
(ex: mise en œuvre, réalisation, gestion)"
  },
  "R3": {
    "name": "verbe_direct",
    "effect": "remplacement des noms abstraits par verbes d'action concrets"
  },
  "S4": {
    "name": "superflu",
    "effect": "ajout d'éléments non nécessaires à la compréhension du fait
(précisions, reformulations, contexte inutile)"
  },
  "R4": {
    "name": "sans_superflu",
    "effect": "suppression de tout élément non essentiel, conservation du fait brut"
  },
  "S5": {
    "name": "syntaxe_complexe",

```

```

    "effect": "structure avec subordination ou enchâssement de propositions"
  },
  "R5": {
    "name": "syntaxe_simple",
    "effect": "une seule proposition principale sans subordination"
  },
  "S6": {
    "name": "sujet_eloigne",
    "effect": "sujet absent, implicite ou placé après le verbe principal"
  },
  "R6": {
    "name": "sujet_initial",
    "effect": "sujet explicite placé en début de phrase avant l'action"
  },
  "S7": {
    "name": "multi_idees",
    "effect": "au moins deux actions ou informations distinctes dans la même phrase"
  },
  "R7": {
    "name": "idee_unique",
    "effect": "une seule action ou information par phrase"
  },
  "S8": {
    "name": "connecteurs_bureaucratiques",
    "effect": "usage de connecteurs institutionnels
    (dans le cadre de, afin de, en vue de)"
  },
  "R8": {
    "name": "sans_connecteur",
    "effect": "absence totale de connecteurs, formulation directe"
  },
  "S9": {
    "name": "redondance",
    "effect": "répétition d'une même information sous plusieurs formes"
  },
  "R9": {
    "name": "non_redondant",
    "effect": "chaque information apparaît une seule fois"
  },
  "S10": {
    "name": "justification",
    "effect": "ajout d'une explication ou justification non nécessaire au fait"
  },
  "R10": {
    "name": "sans_justification",
    "effect": "aucune justification, uniquement le fait brut"
  },
  },
  "INFORMATION": {
    "I1": {
      "name": "dilution",
      "effect": "ajout de mots sans information nouvelle,
      augmentation de longueur sans gain sémantique"
    },
  },

```

```

    "J1": {
      "name": "densite",
      "effect": "chaque segment lexical apporte une information nouvelle et utile"
    },
    "I2": {
      "name": "retard_info",
      "effect": "information principale apparaît après les premiers mots de la phrase"
    },
    "J2": {
      "name": "info_immediate",
      "effect": "information principale placée dès le début de la phrase"
    },
    "I3": {
      "name": "contexte_avant_action",
      "effect": "phrase commence par contexte ou cadrage avant l'action principale"
    },
    "J3": {
      "name": "action_directe",
      "effect": "phrase commence directement par l'action principale"
    },
    "I4": {
      "name": "implicite",
      "effect": "au moins un élément nécessaire à la compréhension est sous-entendu"
    },
    "J4": {
      "name": "explicite",
      "effect": "tous les éléments nécessaires sont formulés sans ambiguïté"
    },
    "I5": {
      "name": "fragmentation",
      "effect": "informations dispersées dans la phrase, ordre non compact"
    },
    "J5": {
      "name": "compacte",
      "effect": "informations regroupées de manière linéaire et compacte"
    }
  }
}

```

C Additional empirical analyses and diagnostics

We complement the main analysis by examining the empirical structure of the latent space through the behavior of the score $\text{Score}_D^{\max}$. This quantity captures the most extreme alignment of a document with respect to the contrastive objective, and therefore provides a sensitivity measure to highly discriminative directions in the embedding space.

A first indication of the structure of this space is given by the correlation between different summary statistics of the score distribution. As shown by the correlation matrix below, mean values, upper quantiles (notably $p90$), and maximal scores exhibit strong dependencies:

$$\begin{bmatrix} 1.00 & 0.68 & 0.06 & 0.84 \\ 0.68 & 1.00 & 0.59 & 0.86 \\ 0.06 & 0.59 & 1.00 & 0.51 \\ 0.84 & 0.86 & 0.51 & 1.00 \end{bmatrix}$$

This structure reveals that $\text{Score}_D^{\max}$ is not an isolated extreme statistic, but is tightly coupled with upper-tail behavior such as $p90$, indicating that extreme alignments are systematically related to the overall distribution of scores. In other words, documents that achieve high maximal alignment also tend to occupy globally high regions of the latent space. This suggests a strongly skewed, heavy-tailed organization of representations, where a small subset of documents concentrates most of the discriminative mass.

This interpretation is reinforced by individual-level observations. The highest-scoring documents correspond to highly structured linguistic instances that align strongly with dominant directions of the embedding space:

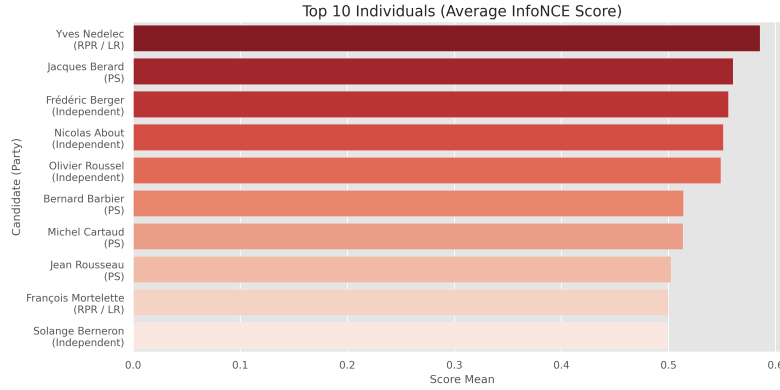


Figure 4: Top 10 highest-scoring individuals in the InfoNCE space according to $\text{Score}_D^{\max}$.

These points do not appear as random outliers but rather as coherent attractors of the representation space, suggesting the presence of localized concentration regions induced by the contrastive objective.

We further examine whether non-discursive variables contribute to the structure captured by $\text{Score}_D^{\max}$. The distributions across gender and age groups show no meaningful separation:

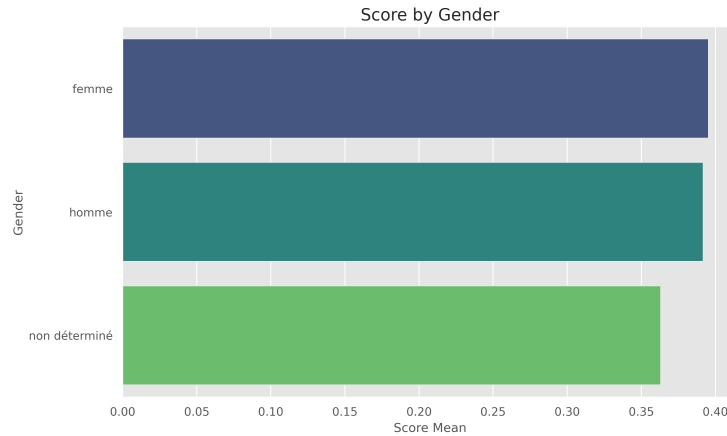


Figure 5: Score distribution by gender. No significant structural separation is observed.

This absence of separation indicates that the geometry induced by $\text{Score}_D^{\max}$ is primarily driven by linguistic and discursive structure rather than demographic metadata, reinforcing the interpretability of the space as a language-induced representation rather than a social embedding.

Finally, the dataset itself consists of 12,746 documents spanning the period 1981–1993, distributed across nine political categories with strong class imbalance. The largest groups (PS, Independent, PCF, RPR/LR, Front National) dominate the corpus, while extreme categories are sparsely represented.

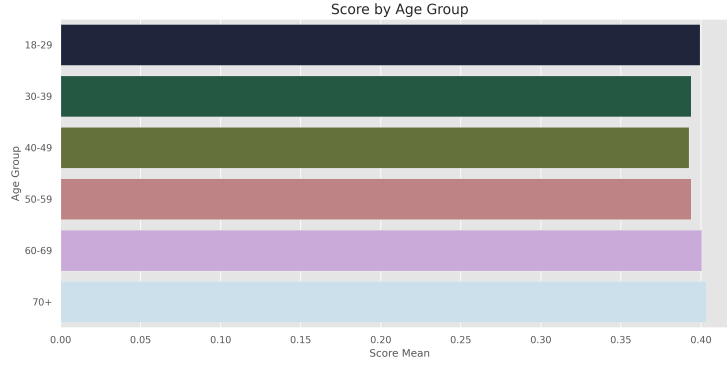


Figure 6: Score distribution by age group. Weak or negligible structure is observed.

Missing values are concentrated in demographic variables (notably age), but do not affect the computation of scores or the structure of the latent space.

Overall, these diagnostics confirm that the latent geometry induced by the contrastive framework is characterized by (i) strong coupling between mean and extreme statistics, (ii) localized concentration effects at the level of individual documents, and (iii) robustness to non-discursive covariates. The score $\text{Score}_D^{\max}$ thus behaves as a reliable probe of extreme alignment structure in the learned representation space.